

HDI Predictor Documentation

No. of Functional Features Included in the Solution

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Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor - Human Development Index Classification System
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Functional Features Overview

S.No	Feature Name	Feature Description	Module / Component	Status (Done/In Progress/Pending)	Marks Contribution
1	Indicator Input Form	Web form to enter life expectancy, mean/expected schooling years, and GNI per capita	Frontend	Done	High
2	Input Validation	Ensures numeric, in-range values before prediction	Backend	Done	High
3	HDI Tier Classification	Predicts Very High / High / Medium / Low using RandomForestClassifier	ML Core	Done	High
4	HDI Score Estimation	Returns the underlying numeric HDI estimate alongside the tier	ML Core	Done	Medium
5	Feature Importance Summary	Shows which indicator most influenced the prediction	ML Core	Done	Medium
6	Result Rendering	Displays tier, score, and explanation on the results page	Frontend	Done	Medium
7	Model Persistence	Loads a pre-trained, serialized model (model.pkl) at startup	Backend	Done	High
8	Docker Packaging	Full app containerized for one-command, reproducible deployment	DevOps	Done	High

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Indicator Input Form, Result Rendering
2	Backend / Logic	2	Input Validation, Model Persistence
3	Database / Storage	1	Serialized model & CSV dataset storage
4	ML / Core Algorithm	2	HDI Tier Classification, Feature Importance Summary
5	Deployment	1	Docker Packaging